

PRINT NAME: _____ ID#: _____

- Show your work. Answers with no or incorrect steps will not receive partial credit.
- The exam is self-explanatory. The instructor will not interpret the questions.

1. (4 points) Let $(X, \|\cdot\|)$ be a Banach space and X_0 a linear subspace of X . Suppose there exists $c \in (0, 1)$ such that

$$c\|y\| \geq \inf_{x \in X_0} \|y - x\|$$

for all $y \in X$. Show that X_0 is dense in X .

Proof. If not, then $X \setminus \overline{X_0}$ is nonempty. By Riesz theorem, for $\epsilon = \frac{1-c}{2} > 0$, there exists $y \in X \setminus \overline{X_0}$, such that $\|y\| = 1$ and $\|y - x\| \geq 1 - \epsilon$ for all $x \in \overline{X_0}$. Hence

$$1 - \epsilon \leq \inf_{x \in X_0} \|y - x\| \leq \inf_{x \in X_0} \|y - x\| \leq c\|y\| = c = 1 - 2\epsilon,$$

which is a contradiction. $\square$

2. (4 points) Let (X, d) be a complete metric space and E is a closed subset of X . Show that E is sequentially precompact if and only if for any $\epsilon > 0$ there exists a sequentially precompact ϵ -net of E .

Proof. ($\Rightarrow$) Since E is sequentially precompact, we know it is totally bounded, and hence for any $\epsilon > 0$, there exists a finite ϵ -net N^ϵ of E . Note that N^ϵ is compact and hence sequentially precompact.

($\Leftarrow$) For any $\epsilon > 0$, there exists a sequentially precompact $(\epsilon/2)$ -net $M^{\epsilon/2}$. Hence $M^{\epsilon/2}$ is totally bounded, i.e., there exists a finite $(\epsilon/2)$ -net $N^{\epsilon/2}$. Hence $N^{\epsilon/2}$ is a finite ϵ -net of E . Hence E is totally bounded and thus sequentially precompact. $\square$

3. (4 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a mapping. Suppose $T^n : X \rightarrow X$ is contractive. Show that T has a unique fixed point in X .

Proof. Since T^n is contractive, we know there exists $\theta \in (0, 1)$ such that $d(T^n x, T^n x') \leq \theta d(x, x')$ for any $x, x' \in X$. By Banach fixed point theorem, there exists a unique $x \in X$ such that $T^n(x) = x$. So $T(x) = T(T^n(x)) = T^n(T(x))$, which implies that $T(x)$ is a fixed point of T^n , and thus $T(x) = x$ since T^n only has one fixed point. Note that, if $T(x) = x$, then $T^n(x) = x$. Hence the fixed point of T is also unique. $\square$

4. (4 points) Let $(X, \|\cdot\|)$ be a Banach space.

(a) Show that every totally bounded subset E of X is bounded.

(b) Let $l^2 := \{x = (x_1, \dots) : x_i \in \mathbb{R}\}$ with norm $\|x\| = (\sum_{i=1}^{\infty} |x_i|^2)^{1/2}$ for any $x \in l^2$. Show that the set $E := \{e_k : k \in \mathbb{N}\}$, where

$$e_k = (\underbrace{0, \dots, 0}_{k \text{ terms}}, 1, 0, \dots),$$

is bounded but not totally bounded in $(l^2, \|\cdot\|)$.

Proof. (a) If E is totally bounded, then there exists a finite 1-net $\{x_1, \dots, x_n\}$ of E . Therefore, for any $x \in E$, there exists $i \in \{1, \dots, n\}$ such that $\|x_i - x\| \leq 1$. Hence

$$\|x\| \leq \|x - x_i\| + \|x_i\| \leq 1 + \max_{1 \leq i \leq n} \|x_i\| =: M.$$

Hence E is bounded by M .

(b) Note that E is bounded since $\|e_k\| = 1$ for all $k \in \mathbb{N}$. However, E is not sequentially precompact since

$$\|e_k - e_j\| = \sqrt{2}, \quad \forall k \neq j.$$

Thus E is not totally bounded. $\square$

5. (4 points) Let (M, d) be a compact metric space and $C(M) := \{f : M \rightarrow \mathbb{R} : f \text{ is continuous}\}$ with norm $\|f\| := \max_{x \in M} |f(x)|$. Let $E \subset C(M)$. Suppose there exist $c, C > 0$, $\alpha \in (0, 1]$, and $\hat{x} \in M$, such that

$$|f(\hat{x})| \leq C \quad \text{and} \quad |f(x_1) - f(x_2)| \leq c d(x_1, x_2)^\alpha$$

for all $x_1, x_2 \in M$ and $f \in E$. Show that E is sequentially precompact.

Proof. For any $\epsilon > 0$, let $\delta = (\epsilon/c)^{1/\alpha} > 0$, then

$$|f(x_1) - f(x_2)| \leq c d(x_1, x_2)^\alpha \leq \epsilon$$

for all $x_1, x_2 \in M$ satisfying $d(x_1, x_2) < \delta$. Hence E is equicontinuous.

Since M is compact, we know it is bounded and hence there exists $D > 0$ such that $d(x_1, x_2) \leq D$ for all $x_1, x_2 \in M$. Therefore

$$|f(x)| \leq |f(x) - f(\hat{x})| + |f(\hat{x})| \leq cD^\alpha + C < \infty$$

for all $f \in E$ and $x \in M$. Hence E is uniformly bounded.

By Arzela-Ascoli theorem, we know E is sequentially precompact in $C(M)$. □